Student Dropout Prevention Using Machine Learning Techniques

1st Jovin Louie
Department of Statistics
University of California, Davis
Davis, California, United States of America
jovlouie@ucdavis.edu

2nd Cesar Aguirre
Department of Computer Science
University of California, Davis
Davis, California, United States of America
cesaguirre@ucdavis.edu

3rd Joey Suen
Department of Statistics
University of California, Davis
Davis, California, United States of America
jtsuen@ucdavis.edu

4th Jessica Zhao
Department of Statistics
University of California, Davis
Davis, California, United States of America
jszhao@ucdavis.edu

5th Eli Seligman
Department of Statistics
University of California, Davis
Davis, California, United States of America
erseligman@ucdavis.edu

***Abstract*—Student dropout in higher education remains a significant challenge for both institutions and students. This project investigates whether machine learning models can effectively identify students at risk of dropping out early enough to allow meaningful intervention or allow for better usage of institutional resources. Using the Predict Students' Dropout and Academic Success dataset from the Polytechnic Institute of Portalegre, we trained and evaluated three classification models: Logistic Regression, Random Forest, and XGBoost. The models were built using demographic, financial, and early academic performance features, with recall emphasized as a primary evaluation metric to prioritize identifying at-risk students. Results show that each model exhibits different strengths depending on institutional priorities, such as maximizing recall or reducing false positives. Feature analysis also highlights the importance of both academic progress and financial stability in predicting student outcomes. While the models demonstrate promising predictive performance, the dataset's origin might limit the ability to generalize for other universities around the globe. Future work should focus on validating similar models across different universities and incorporating additional behavioral or engagement-based features to improve robustness.**

***Index Terms*—Machine learning, student dropout prediction, classification, logistic regression, random forest, XGBoost, higher education analysis.**

I. Introduction

Student dropout and academic underperformance in higher education has always been a significant challenge for higher education institutions worldwide. Universities invest time and resources into supporting students and creating quality programs in various academic fields, yet many students still drop out before finishing their degree. High dropout rates not only impact universities, but also impact student success and society as a whole, since students who leave early may experience hardships when searching for employment opportunities or increased risk of future financial instability. As more students aim to make higher education their goal, addressing challenges of student retention becomes an essential priority for educators and policymakers. In addition, improving student retention is also tied to the reputation of the institution, financial sustainability, and the overall effectiveness of the education in the actual workforce.

Recent technological developments in the area of machine learning (ML) have made it possible to address many complex issues with large amounts of data, including in higher education. ML is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions without being explicitly programmed for each task. Because ML techniques can process large datasets and identify complex relationships between variables, they are increasingly used to support decision making in various fields.

Since one of the biggest problems in higher education remains student dropout and academic underperformance, ML provides for a powerful tool that should be leveraged. ML can create predictive models that have the ability to analyze historical student data and estimate the likelihood that a student may struggle or leave their program. Traditionally, analysis of student dropout rates often relied on manual review of minimal student records or simple statistical analyses, which is insufficient when dealing with such a large-scale problem. Now, institutions in higher education collect data from students, such as various demographic information and socioeconomic status. Additionally, macroeconomic data such as GDP and inflation

rate is also made publicly available. ML can be used to analyze this data by creating a model that can be used to identify early risk identification. These models are capable of detecting relationships between variables that may not show through traditional analysis methods. Predictions from the model can be used to help aid both students and universities in making smart decisions. For example, institutions can design academic advising programs or financial support systems to assist students who are identified as high risk. Previous research has already demonstrated the effectiveness of machine learning in predicting student dropout and academic performance, which will be discussed further in the literature review section.

In this project, we explore the use of machine learning techniques to identify students who may be at risk of dropping out by developing 3 predictive models and evaluating their performance using classification metrics. Our goal is to create a binary classification model capable of early identification of university students at high risk of dropping out based on data such as academic path, demographics, and socioeconomic factors. This information would be greatly useful as students can be given the support that they need to avoid dropping out and institutions can also allocate resources accordingly. Ultimately, this approach demonstrates how ML can contribute to improving student outcomes and strengthening the effectiveness of higher education systems.

RQ1: Which demographic, socioeconomic, and academic features are the most significant predictors of student dropout in higher education?

RQ2: How accurately can machine learning models predict student dropout, and which model (Logistic Regression, Random Forest, or XGBoost) performs best according to evaluation metrics such as accuracy, precision, recall, F1-score, MCC, and ROC-AUC?

RQ3: How can results from machine learning models be used to support early identification of at-risk students and guide institutional decisions to improve student retention?

II. Literature Review

The first article reviewed is titled "Predicting Student Dropout and Academic Success" by Realinho et al [1]. The authors formulated a curated dataset on the topic of predicting student dropout and academic success by compiling data from different disjointed data sources that includes demographic (marital status, nationality, displaced/not displaced, gender, and age at enrollment), socioeconomic (mother and father's qualification), macroeconomic (unemployment & inflation rate, GDP), and academic (enrollment) information. The dataset has 4,424 records of students enrolled in the Polytechnic Institute of Portalegre from the academic years 2008/2009 to 2018/2019, includes 17 undergraduate degrees in different fields, and consists of 35 predictive attributes with no missing values. Similar to our project, this problem was also formulated as a 3-category classification task, in which there is a strong imbalance towards the "Graduate" class. The authors encouraged its dataset users to address it with either a data-level approach (sampling technique such as Synthetic Minority Over Sampling Technique) or an algorithm-level approach (a machine learning algorithm that already incorporates balancing steps).

The second article reviewed is titled "A novel AI-driven model for student dropout risk analysis with explainable AI insights" by Mustofa et al [2]. The authors proposed a comprehensive machine-learning and explainable AI-based framework for predicting student dropout risk while also identifying the key contributing factors behind such predictions. The study utilizes a publicly available higher education dataset containing 4,424 student records with 34 predictive attributes spanning demographic (ex: gender, age at enrollment, nationality), socioeconomic (ex: parental occupation, scholarship status, debt status), academic, and macroeconomic indicators (ex: unemployment rate, inflation rate, GDP). The original dataset is formulated as a multi-class classification problem and exhibited a notable class imbalance. To address this, the authors merged the "Graduate" and "Enrolled" categories into a single "Non-dropout" category and applied an upsampling strategy to balance the dataset. Feature selection was conducted using Recursive Feature Elimination with Cross-Validation to retain the most important predictors. The study introduced a hybrid model (Hybrid Logistic Regression and Neural Network) that combines the interpretability of logistic regression with the non-linear pattern learning capability of the artificial neural networks. The proposed model achieved an accuracy of 96% and outperformed traditional ML models. The authors finally employed explainable-AI techniques to interpret model predictions which highlighted critical dropout risk factors such as gender, displacement status, debt status, thus providing actionable insights for early intervention and student retention strategies.

III. Dataset Description and Exploratory Data Analysis

We are utilizing the "Predict Students' Dropout and Academic Success" dataset available from the UC Irvine Machine Learning Repository. This dataset encompasses 4,424 students and 36 features across demographic, socio-economic, and academic dimensions, aimed at predicting outcomes of Dropout, Enrolled, or Graduate. The dataset was created in an attempt to help reduce the academic dropout and failure rate in higher education by identifying at-risk students at earlier stages in their academic paths. The data was sourced from the Polytechnic Institute of Portalegre in Portugal and hosted by the UC Irvine Machine Learning Repository.

Analysis of discrete features revealed significant positive skewness in Application Order (1.66) and Age at Enrollment (1.29), while Curricular Units Approved showed a slight negative skew (-0.56). Utilizing the Interquartile Range (IQR) method, we identified outlier rates of approximately 12% in Curricular Units and Application Order, alongside a 10% outlier rate for Age. These anomalies highlight distinct subpopulations, such as transfer students, adult learners, and those placed in secondary-choice courses.

To analyze continuous variables, a heatmap (Figure 1) was generated by mapping the target outcomes to a -1 to 1 numeric scale. This revealed that first and second-semester grades are the strongest predictors of graduation, maintaining a high internal correlation of 0.84. While admission and qualification grades showed a moderate link (0.58), macro-economic factors like GDP and inflation had negligible impacts on student outcomes. This suggests that internal academic performance far outweighs external economic conditions in determining a student's likelihood of success.

Fig. 1. Correlation Analysis Heatmap: Continuous Academic & Economic Factors

Binary and categorical analyses identified financial stability as the primary driver of student retention. Although a vast majority of the student body (88.1%) maintains up-to-date tuition fees, the impact of falling behind is severe. Students with up-to-date accounts achieve a 56.0% graduation rate, contrasted with a 5.5% rate for those in arrears, representing a massive 50.5 percentage point success gap. Furthermore, the 24.8% of students who secure scholarships experience a 76.0% graduation rate, significantly outperforming the 41.3% graduation rate of non-holders. Conversely, the 11.4% of students classified as debtors face a severe disadvantage, graduating at a rate of just 20.1% compared to 53.8% for their non-debtor peers.

Significant demographic and social trends also emerged from the data, highlighting the vulnerabilities of specific student groups. Gender plays a notable role, with female students graduating at a rate of 57.9%, which is 22.7% higher than the 35.2% graduation rate observed for male students. Marital status also heavily influences student outcomes. As seen in Table I, single students experience a baseline dropout rate of 30.2%. In contrast, married and divorced students face a considerably higher risk, exhibiting dropout rates of 47.2% and 46.1%, respectively. This retention risk peaks dramatically for students legally separated, who face a staggering 66.6% dropout rate and only a 16.6% chance of graduation. Interestingly, "displaced" students, who make up 54.8% of the dataset, demonstrate resilience, showing a 10.3 percentage point higher graduation rate than non-displaced students.

TABLE I
MARITAL STATUS IMPACT ON GRADUATION AND DROPOUT RATES

ID	Marital Status	Dropout Rate	Graduate Rate
1	*Single*	30.2%	**51.4%**
2	*Married*	47.2%	39.0%
4	*Divorced*	46.1%	36.2%
5	*Facto Union*	44.0%	44.0%
6	*Legally Separated*	**66.6%**	**16.6%**

Beyond financial and demographic factors, enrollment conditions and specific sub-populations provide further context on student success. A significant majority of the student body (89.1%) is enrolled with standard daytime attendance, which correlates with a 51.0% graduation rate, compared to a lower 41.6% rate for those in evening or non-standard attendance (a 9.3 percentage point impact difference). Meanwhile, niche sub-populations such as international students (2.5% of the dataset) and students with educational special needs (1.2%) represent very small margins of the cohort. Interestingly, these groups exhibit relatively neutral impacts on graduation likelihood; international students graduate at a rate of 49.1% (compared to 50.0% for domestic students), and students with special needs graduate at a rate of 45.1%

Finally, academic enrollment choices and curricular distributions reveal stark contrasts in student success. Course selection is highly indicative of retention; for instance as seen in Table II, Nursing students (Course 9500) achieve a peak graduation rate of 71.5% and a low dropout rate of 15.4%. On the other end of the spectrum, students enrolled in Management (Course 9147) struggle significantly, showing a 35.2% dropout rate and only a 36.3% graduation rate. From a data distribution perspective, academic features such as "Curricular units 1st sem (credited)" and "Age at enrollment" exhibited high outlier percentages (13.04% and 9.97%, respectively) alongside significant positive skewness (0.90 and 1.29). This indicates that while the majority of the population fits a standard academic profile, a distinct subgroup requires non-traditional academic pacing, further complicating retention efforts.

TABLE II
COURSE IMPACT ON GRADUATION AND DROPOUT RATES

Course ID	Course Name	Dropout Rate	Graduate Rate
9500	*Nursing*	15.4%	**71.5%**
9147	*Management*	**35.2%**	36.3%
9238	*Social Service*	18.3%	69.8%
9085	*Veterinary Nursing*	26.7%	51.0%
9773	*Journalism and Communication*	30.5%	59.2%

While marital status and course selection show extreme variances in student outcomes, parental occupation categories demonstrate more moderate impacts. An analysis of the top

five most frequent Mother's and Father's occupations reveals graduation rates that generally fluctuate between 43% and 55%. For instance, the most frequent Father's occupation (Unskilled Workers) yields a 51.8% graduation rate and a 31.9% dropout rate. However, Administrative Staff presents the highest risk among the top father occupations, exhibiting a 36.0% dropout rate and dropping the graduation rate to 44.8%. Similarly, Mother's occupation Specialists in Intellectual and Scientific Activities acts as a localized risk factor, yielding a lower graduation rate of 43.3% paired with a 32.0% dropout rate.

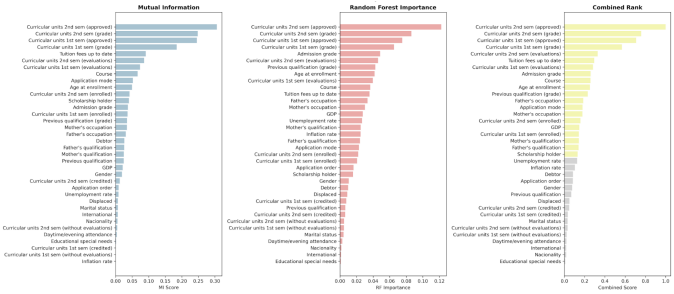

Fig. 2. Feature Selection

IV. Methodology

A. Feature Selection and Dataset Preparation

In order to obtain the proper features for our models we decided to perform feature selection. Two metrics were used to evaluate the features: mutual information and Random Forest importance. We used both metrics to gain a more comprehensive understanding of which features are significant. Note that we used the original dataset, which is label encoded for the categorical variables, in order to be able to compare all features at the same time.

Mutual information measures the mutual dependence between each feature and the target (the final classification) using the joint and marginal probabilities. In other words, it measures how much a feature (in isolation) reduces entropy. Thus, this metric was used to first compute the strength of relationship between each isolated feature and the target, in the context of the entire dataset.

For Random Forest importance, 300 different decision trees were generated, and the information gain for each split was tracked. Then, the average information gain for each feature was computed. Like mutual information, Random Forest importance also measures how much each feature reduces entropy. However, it measures this in the context of other decision tree splits, allowing Random Forest importance to also capture interaction effects.

Finally, to select the features used in our model, we generated a combined score that averages the mutual information and Random Forest importance, and ranked them. The selected features, in yellow, are the features that account for 90% of the total combined score. Since our main metric for our model is recall, which is discussed later, we chose 90% because we want to include more features and avoid removing any information that helps us identify a potential dropout.

In order to address the lack of dropouts within our dataset, we decided to take two specific measures. We modified our cost function to penalize false negatives (i.e. missing a dropout) more than false positives (misclassifying a non-dropout). Another strategy we have come up to address this problem is simply testing different threshold values for our classifier. Moving between different threshold values could have the effect of hopefully increasing the performance and recall of our classifier models.

We split the dataset into a 60-25-15 training, testing, and demonstration split to ensure that the models could be trained effectively while still reserving sufficient data for unbiased evaluation and presentation. The training set (60%) was used to fit the models and perform cross-validation during development. The test set (25%) was kept separate during training and used only to evaluate final model performance on unseen data. The remaining 15% of the data was reserved as a demonstration set for the in-class presentation demonstration on further unseen data. This separation ensures that the reported metrics better reflect how the models would perform when applied to new student records rather than data they were trained on.

Before splitting, the dataset was randomly shuffled using sklearn's train_test_split, and stratified sampling was applied to ensure that the proportion of dropout and non-dropout students remained consistent across the training, test, and demo subsets. The raw dataset already contained numerically encoded features and did not include any missing values, so minimal preprocessing was required prior to training. The target variable was converted into a binary label representing dropout versus non-dropout to match the binary classification objective of our models. Maintaining these separate dataset partitions helped prevent data leakage and ensured that the reported evaluation metrics more accurately reflect how the models would perform on unseen student data.

B. Model Development

In order to effectively capture the patterns from our data, we decided to train and test three models on our dataset. In order to prevent overfitting, we decided to apply 5-fold stratified cross-validation while training. In particular, we chose to use stratified cross-validation to ensure that the class distribution of our target (dropouts vs. non-dropouts) remained consistent across folds, which was critical due to the imbalance nature of our dataset. Our cross-validation monitored precision, recall, and F1 during testing.

A **Logistic Regression** was developed as a baseline due to its simplicity and interpretability, as well as to inform future decisions concerning model development in the case that decision boundaries were not linearly separable. Several ratio-based features were engineered in order to add complexity to the linear regression model, allowing the linear coefficients to capture real-world academic relationships.

The `pass_rates` feature represents the proportion of registered courses passed by a student within first, second, and

across both semesters. `average_gpa` is the mean GPA value across both semesters. `academic_improvement` captures the difference between second and first semester grades. Finally, `evaluation_efficiency` is the proportion of curricular approved units out of the total unit evaluations. All features were transformed by Z-score normalization due to the impact feature scaling has on logistic regression. Hyperparameter tuning was conducted using `GridSearchCV`, performing an exhaustive search over all combinations of possible hyperparameters to find the optimal values that maximize the F1-score. The following hyperparameters were tuned:

The regularization strength was controlled via `C` (values: 0.001, 0.01, 0.1, 1, 10, 100), and the penalty term via `penalty` (`None`, `l2`, `l1`, `elasticnet`). The optimization algorithm was set through `solver` (`liblinear`, `saga`), and class imbalance was addressed via `class_weight` (`none`, `balanced` (adjusting weights inversely proportional to class frequency in the input data)).

Note: 60 of the total 480 possible fits (5 folds across 96 candidate hyperparameter combinations) fail, as the `saga` solver does not support the `elasticnet` penalty.

The optimal hyperparameter configuration identified by `GridSearchCV` was `C` $= 0.01$, `penalty` $=$ `l2`, `solver` $=$ `liblinear`, and `class_weight` $=$ `balanced`.

Our second model was **Random Forest**, which was chosen because of its power in handling complex, noisy tabular data and its ability to mitigate overfitting via ensembling, as well as its ability to capture deeper statistical patterns. To address the class imbalance between dropouts and non-dropouts once again, we used balanced class weighting. Our Random Forest used an ensemble of 300 decision trees and aggregated the predictions through majority voting. Unlike Logistic Regression, we did not make any modifications to our features, since Random Forest is robust to the scale and distribution of individual features.

Our third model was **XGBoost**, which was chosen because of its ability to mitigate overfitting (similar to Random Forest) as well as being able to handle nonlinear relationships and feature interactions effectively. The ratio of dropouts to non-dropouts was entered as a parameter to control the balance of positive and negative weights and account for our class imbalance. Additional regularization was applied by randomly subsampling 80% of the data and 80% of the features to train each tree, reducing overfitting. The model was trained with 200 boosting rounds with a learning rate of 0.1 and a maximum tree depth of 5, which also helps prevent overfitting and allows for better computational efficiency. Since our task was binary classification, and XGBoost outputs probabilities before turning them into classifications, log loss was the natural choice for our model's evaluation metric.

For all three models, we performed evaluation by computing accuracy, precision, recall, F1, MCC, and ROC-AUC.

V. Results and Evaluation

A. Results

As seen in Figure 6, models converge on approved second semester units as the most significant predictive signal, alongside a shared set of top features. Curricular units in the first semester, up-to-date tuition fees, grades in second semester, and age of enrollment also ranked among the ten most significant features for each of our models, although variance in ranks indicates model-specific sensitivities.

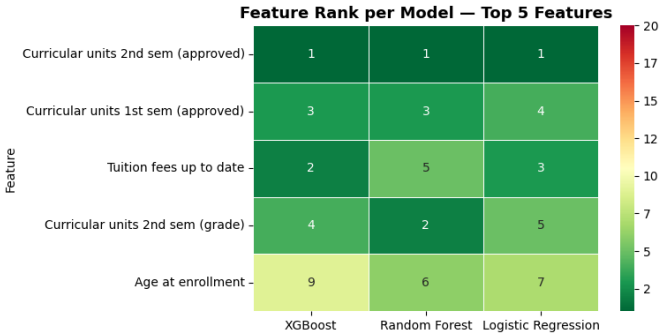

Fig. 3. Feature Agreement

Overall, these most significant features tend to fall within one of three categories: academic momentum, financial stability, and student profile. Approved second-semester units indicate that academic engagement and progression are a strong indicator for student retention. Keeping tuition fees up to date may reflect financial stability, a well-documented predictor of dropout. Age at enrollment could relate to specific life circumstances, as older students may have competing responsibilities (i.e., work and family) that may increase their risk of dropout.

There was a large degree of agreement in what the least informative features were to each of our models, observed in the figure below.

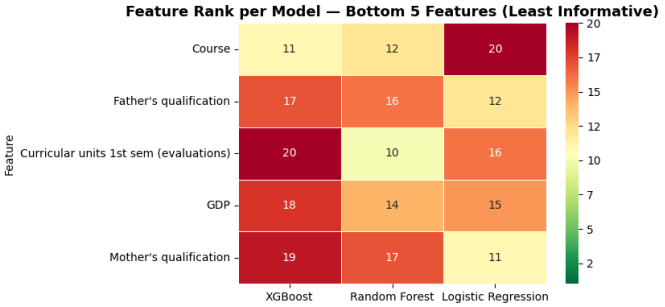

Fig. 4. Feature Agreement (Least important)

GDP ranks low across all three models, an indication that macroeconomic conditions had little predictive power within this dataset, likely caused by the fact that all of our students were from Portugal. Also, parental qualifications were consistently unimportant in all 3 models, which may push back

against some socioeconomic dropout literature. Course showed the most disagreement ranking 11th within the XGBoost model but last by Logistic Regression, indicating some model-specific sensitivity to categorical program enrollment.

B. Evaluation

Among our six metrics, recall was regarded with the greatest practical importance, since missing a student at risk of dropout has greater real-world consequences than incorrectly flagging a student. Our Logistic Regression model achieved the highest recall score at 0.811, correctly identifying over 81% of actual dropouts. However, the trade-off was that Logistic Regression had the worst precision, meaning over 28% flagged students would not have actually dropped out. Its ROC-AUC of 0.907 and MCC of 0.642 are high, though, indicating a strong classification ability nonetheless.

TABLE III
PERFORMANCE COMPARISON OF LOGISTIC REGRESSION (LR), RANDOM FOREST (RF), AND XGBOOST (XGB) MODELS

Model	LR	RF	XGB
Accuracy	0.837	**0.863**	0.860
Precision	0.718	**0.84**	0.791
Recall	**0.811**	0.708	0.767
F1	0.762	0.768	**0.779**
MCC	0.6423	**0.676**	**0.676**
ROC-AUC	0.907	0.9113	**0.9170**

Random Forest and XGBoost also offered different trade-offs. Random Forest achieved the highest accuracy (0.863) and precision (0.84), and thus had the most reliable dropout predictions. However, it had the worst recall of the three at 0.708, meaning that it missed nearly 30% of actual dropouts. XGBoost, on the other hand, performed the most consistently across the board, with solid accuracy, precision, and recall, and the highest F1, MCC, and ROC-AUC scores. Therefore, out of the three models, XGBoost had the best overall model quality and general classification ability, since it excelled in the more holistic metrics that summarize model quality.

Overall, none of the three models stood out as the "best" model. All three models had strong discrimination abilities, as can be seen by the relatively high MCC, all achieved over 0.64, and ROC-AUC, all achieved over 0.9, scores. This validates that our selected features carry significant predictive signal. Furthermore, each model offered its own specialty. Logistic Regression was the best at identifying actual dropouts, Random Forest was the best at predicting dropouts, and XGBoost offered the best balance of prediction and model quality. Ultimately, the choice of model is dependent on an educational institution's resources, capacity, and priorities, which is further discussed in the subsequent section.

Figure 5 emphasizes that all three models are viable classifiers, outperforming the random baseline, however the question remains to determine which model best aligns with an institution's priorities.

Between recall values 0.2 and 0.6, the three models are nearly indistinguishable, indicating that institutions with conservative "flagging" thresholds, where only the most at-risk

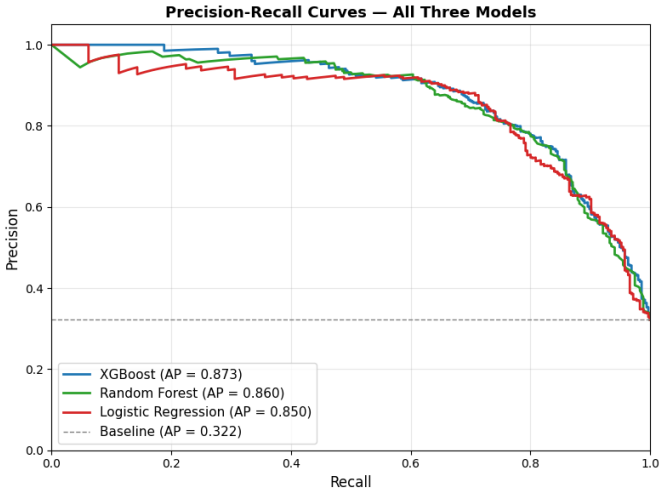

Fig. 5. Precision-Recall Curves of All

students are identified, would see comparable performance between all three models. We observe meaningful divergence at recall values above 0.8, the region most relevant to our objective of missing dropouts. Logistic regression extends the furthest along the recall axis, but at a steeper precision cost, meaning more flagged students would not have actually dropped out. In contrast, the other two models maintain high precision in this high-recall region, again offering more reliable flags at the cost of missing true dropouts. Ultimately, no single model is the holistic "best" choice, and right selection depends on an institution's intervention capacity. A well-resourced advising program that can follow up with a larger flagged population may favor the higher recall from Logistic Regression, but a program with limited capacity may prefer XGBoost's more precise flags.

Concerning future model implementation, Figure 6 below indicate that XGBoost and Random Forest plots peak early, to the left of 0.5, indicating that they could benefit from lenient flagging, catching more students before dropping precision too steeply.

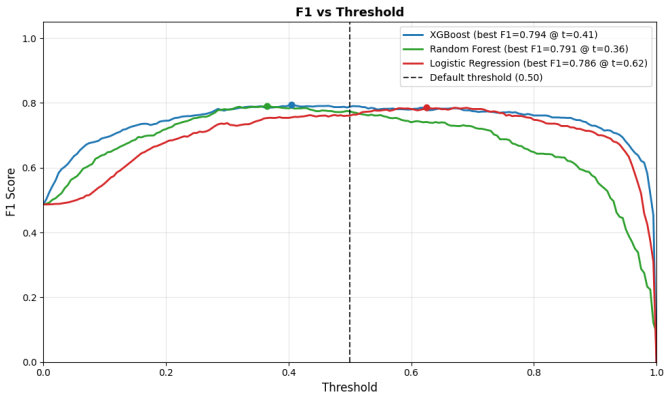

Fig. 6. Threshold Sweep

Interestingly, Logistic Regression would benefit from a

more strict threshold value to maximize F1 score, aligning with the Figure 5's finding that Logistic Regression has a high recall but weaker precision. Previous statements on institutions receiving comparable performance from all three models is also supported by the threshold sweep plot, given that all three curves overlay and plateau between the threshold values of 0.3 and 0.6. We also see a stark Random Forest drop-off after 0.6, which is consistent with its high precision, low recall profile.

VI. Conclusion and Discussion

This project set out to explore whether machine learning could realistically be used to flag students at risk of dropping out before it's too late to help them. Using a dataset from the Polytechnic Institute of Portalegre, we trained three models (Logistic Regression, Random Forest, and XGBoost) on demographic, financial, and academic data collected early in students' enrollment. Since the whole point of a system like this is to catch at-risk students in time for intervention, recall was treated as one of our main metrics throughout evaluation.

As discussed in the results, the three models each had distinct strengths depending on which metrics are prioritized. The takeaway isn't that one model is simply "best", but that the right choice really depends on what a given institution can realistically act on. A school with limited advising resources might favor precision to avoid overwhelming staff with false positives, while one focused on reaching every at-risk student possible might accept a higher false positive rate to maximize recall. The feature analysis similarly reinforced what the results section covers in more detail: both academic progress and financial stability emerged as meaningful signals worth monitoring together rather than in isolation.

One important limitation to be aware of is that this dataset comes from a single Portuguese polytechnic institution, and that context shapes the results more than it might initially seem. Portugal's tuition structure, financial aid system, and student population are different enough from, for example, a large public university in the U.S. such that you couldn't just drop these models into another setting and expect the same performance. Some of the predictive relationships we found might hold across institutions, but others are probably specific to this particular environment.

That makes cross-institutional validation a natural next step for anyone looking to build on this work. Testing similar models on data from universities in different countries, or even different types of institutions within the same country, would help clarify which predictors are broadly useful and which are more contextual. It would also likely require rethinking some of the features used here; variables like campus resource utilization, student engagement patterns, or online activity logs might turn out to be more informative in other settings than the administrative data available in this dataset.

Overall, the results are encouraging as a proof of concept. Machine learning can meaningfully identify at-risk students using data that institutions already collect, and the models performed well enough to be worth developing further. But meaningful deployment at scale, across different schools, systems, and student populations, will require further research and models tailored to the environments they are intended to serve.

VII. GitHub Link:

https://github.com/JovinLouie/ECS-171-Project

VIII. Project Roadmap

We were able to complete the main goals outlined in our initial project roadmap from the 1-pager. First, we defined the objective of predicting whether a student would drop out or remain enrolled using demographic, socioeconomic, and academic information from the dataset. The dataset originally contained three outcomes (Dropout, Enrolled, and Graduate), but for the purpose of our study we converted it into a binary classification problem by grouping Enrolled and Graduate together as non-dropout. This allowed us to focus specifically on identifying students who are at risk of dropping out.

Next, we performed exploratory data analysis (EDA) to better understand the structure of the dataset and identify patterns related to student outcomes. During this stage, we examined the distributions of important features, calculated summary statistics, and created visualizations to observe relationships between variables and the target outcome. This helped us identify several key trends, such as the strong connection between academic performance and graduation outcomes, as well as the influence of financial factors like tuition payment status and scholarship availability. EDA also helped us identify skewed variables and potential outliers in features such as application order and age at enrollment.

After gaining a better understanding of the dataset, we moved on to feature selection. We used two different methods to evaluate feature importance: mutual information and Random Forest importance. Mutual information helped measure how strongly each individual feature was related to the target variable, while Random Forest importance captured how useful each feature was within a more complex model that considers interactions between variables. We then combined the results from both methods to rank the features and selected the ones that accounted for most of the overall predictive importance. In addition to the existing dataset features, we also engineered several new academic performance features, such as semester pass rates, overall pass rate, grade improvement, and evaluation efficiency, to better capture patterns in student academic progress.

Once the feature set was finalized, the dataset was prepared for model training. The data was randomly shuffled and split into three subsets: a training set, a test set, and a smaller demonstration set. Stratified sampling was used so that the proportion of dropout and non-dropout students remained consistent across each split. This helped ensure that the models would be trained on representative data while still allowing us to evaluate their performance on unseen data.

We then trained three machine learning models: Logistic Regression, Random Forest, and XGBoost. Logistic Regression was used as a baseline model because it is simple

and easy to interpret. Random Forest was included because it can capture more complex patterns through an ensemble of decision trees. Finally, we used XGBoost, which is a powerful gradient boosting algorithm that can model nonlinear relationships and feature interactions while also including regularization to reduce overfitting. All models were trained using 5-fold stratified cross-validation to maintain balanced class distributions during training.

Finally, the models were evaluated using the test set. We compared their performance using several metrics including accuracy, precision, recall, F1-score, Matthews Correlation Coefficient (MCC), and ROC-AUC. Since the main goal of this project was to identify students who are at risk of dropping out, recall was treated as one of the most important metrics. By comparing these results across models, we were able to see how well each approach performed and which models were more effective at identifying at-risk students.

IX. Assignment of Tasks to Team Members

The introduction and background section was written by Jessica Zhao. She read the published dataset information (including purpose, funding, and dataset preprocessing) from the UCI Machine Learning Repository and reported the relevant information needed to proceed with our project. Additionally, she researched background information from the two published papers included in our literature review as well as other sources on the problem of student dropout and academic underperformance in higher education, and connected this problem to recent developments in ML. In the project presentation, she created and presented the "Problem" slide.

The literature review section was done by Jessica Zhao and Jovin Louie, who each reviewed one previously published paper on the topic of using ML to help the problem of student dropout by analyzing their methodology, summarizing their conclusions, and identifying their differences with the aim of our project.

The dataset and exploratory data analysis for this project was done by Eli Seligman. He provided a comprehensive Dataset Description, where he categorized and defined the demographic, socio-economic, and academic variables to establish a clear context for the study. Then, he conducted an extensive Exploratory Data Analysis (EDA) to uncover key correlations and trends within the data that influence student outcomes.

After EDA, Joey Suen analyzed the features and identified the best features based on mutual information and Random Forest importance.

Model development outline was done by Cesar Aguirre. Logistic Regression was identified as the baseline classifier in this stage, as well as the outlines for Random Forest and XGBoost implementation to observe more complex interactions between features.

Our methodology section was written by Jovin and Joey in order to explain the work for our following model development. For the creation of our 3 models, we decided to divide the work for each model amongst 1-2 people per model while using a common training and testing framework with a 5-fold cross validation.

For our Logistic Regression model, Cesar Aguirre did the baseline model as well as the feature engineering to allow the logistic regression model to capture certain non-linear relationships within the dataset. Hyperparameter tuning for the logistic regression was also done by Cesar. Our Random Forest model was trained and tested by Jovin Louie and Jessica Zhao, who worked together to implement the model and evaluate its performance in order to capture the possible non-linear relationships and to prevent overfitting.

Eli developed an XGBoost classification model; by leveraging this gradient boosting algorithm, he captured complex, non-linear relationships to effectively identify students at risk of dropping out. Joey constructed the front end for the demonstration of our three models, utilizing the Streamlit framework to create an interactive interface that tests the three models on a completely new validation set and visualizes the five evaluation metrics, the ROC curve, and the relative importance of the twenty features used for each model.

References

[1] V. Realinho, J. Machado, L. Baptista, and M. V. Martins, "Predicting Student Dropout and Academic Success," Data, vol. 7, no. 11, p. 146, Nov. 2022, doi: https://doi.org/10.3390/data7110146.

[2] S. Mustofa, Y. R. Emon, S. B. Mamun, S. A. Akhy, and M. T. Ahad, "A novel AI-driven model for student dropout risk analysis with explainable AI insights," Computers and Education: Artificial Intelligence, vol. 8, p. 100352, Dec. 2024, doi: https://doi.org/10.1016/j.caeai.2024.100352.